





PROJECT REPORT ON

Crypto Trading Platform using Machine Learning with AI"

BY

Tushar Gupta

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Certificate

This is to certify that Mr. **Tushar Gupta** is a bonafide student of **School of Computer Applications**, Pimpri Chinchwad University, Sate Maval, Pune has successfully completed the project work as prescribed by **Pimpri Chinchwad University**, in the partial fulfilment of **Master of Computer Applications (MCA) Program** for the Academic Year 2025-26.

The project work, entitled as **Crypto Trading Platform using Machine Learning with AI** has been carried out at the establishment of
“ _____ ”

Roll No.: MCB24251

PRN No.: SCA24801030120

Exam Seat No.: _____

Mr. Sanjay Dandele

Dr. Kamlesh Meshram

Internal Guide

Project Co-ordinator

Head, SOCA

Declaration

I hereby declare that the project work entitled, **Crypto Trading Platform Using Machine Learning With Ai** submitted under guidance of Prof. **Madhuri Dharro**, is my original work completed under the four walls of Maincraft Technology

The Project Report submitted is my own work and has not been duplicated from any other source. I shall be responsible for any unpleasure moment/situation.

Tushar Gupta

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Tushar Gupta

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CHAPTER 1: INTRODUCTION

1.1 Introduction

Cryptocurrency has emerged as one of the most innovative and rapidly growing financial technologies in recent years. It enables secure, decentralized, and transparent digital transactions without the involvement of traditional banking systems. Popular cryptocurrencies such as Bitcoin and Ethereum have gained widespread acceptance, attracting investors, traders, and organizations worldwide.

However, cryptocurrency markets are highly volatile and unpredictable, making trading a challenging task. Prices fluctuate rapidly due to various factors such as market demand, global events, investor sentiment, and technological changes. Traditional trading methods rely heavily on manual analysis, which is time-consuming, error-prone, and often influenced by human emotions like fear and greed.

To overcome these challenges, the integration of **Artificial Intelligence (AI)** and **Machine Learning (ML)** has become essential. AI and ML techniques can analyze large volumes of historical and real-time data, identify patterns, and predict future price trends with improved accuracy. These technologies help traders make informed decisions, reduce risks, and enhance trading efficiency.

This project, titled “**Crypto Trading Platform using Machine Learning with AI**”, aims to develop a smart and user-friendly system that combines modern web technologies with intelligent algorithms. The platform allows users to view real-time cryptocurrency data, analyze market trends, receive AI-based trading recommendations, and manage their portfolios effectively.

The system is designed using modern technologies such as React for frontend development, backend frameworks like Node.js or Python, and Machine Learning libraries for prediction models. It also integrates external APIs to fetch real-time market data and ensures secure user authentication and data handling.

Overall, this project provides a comprehensive solution for cryptocurrency trading by combining **data analytics, AI-driven insights, and user-friendly design**, making trading more efficient, accurate, and accessible for both beginners and experienced users.

1.2 Background / Problem Statement

Background

Cryptocurrency is a digital form of currency that operates on decentralized networks using blockchain technology. Unlike traditional financial systems, it does not rely on central authorities such as banks or governments, making transactions more secure, transparent, and efficient. Over the past decade, cryptocurrencies like Bitcoin and Ethereum have gained global recognition and widespread adoption.

With the increasing popularity of cryptocurrencies, online trading platforms have emerged to facilitate buying, selling, and managing digital assets. These platforms provide users with access to real-time market data, charts, and trading tools. However, due to the highly dynamic nature of the crypto market, analyzing trends and making profitable decisions remains a challenging task.

The rapid growth in the volume of trading data has made manual analysis inefficient and time-consuming. Traders often rely on historical data and basic indicators, which may not always provide accurate insights into future price movements. Additionally, market behavior is influenced by multiple factors such as global economic conditions, technological developments, and investor sentiment.

In this context, the use of **Artificial Intelligence (AI)** and **Machine Learning (ML)** has become increasingly important. These technologies can process large datasets, identify hidden patterns, and generate predictive models for better decision-making. By integrating AI into crypto trading platforms, it becomes possible to enhance accuracy, reduce risks, and improve overall trading efficiency.

This project is based on the concept of combining modern web technologies with AI and ML techniques to create a smart crypto trading platform that provides real-time analysis, predictive insights, and a user-friendly experience.

Problem Statement

The current methods and tools used for flare system design present the following challenges:

1. Lack of AI-based prediction capabilities for accurate decision-making
2. Heavy reliance on manual analysis of market trends
3. Difficulty in processing large volumes of real-time data
4. High risk due to market volatility in cryptocurrencies like Bitcoin and Ethereum
5. Time-consuming analysis leading to delayed decisions
6. Increased chances of human errors and emotional trading
7. Limited support for automated trading recommendations

1.3 Objectives of the Project

The main objective of this project is to develop an intelligent and user-friendly **Crypto Trading Platform using Machine Learning with AI** that enhances trading efficiency and decision-making.

Specific Objectives

- To develop a secure and scalable crypto trading platform for users
- To implement Machine Learning algorithms for predicting price trends of cryptocurrencies like Bitcoin and Ethereum
- To provide real-time market data and analysis using external APIs
- To reduce human errors and emotional bias in trading decisions
- To design a user-friendly interface for both beginners and experienced traders
- To enable portfolio management and transaction tracking
- To provide AI-based trading recommendations for better decision-making

- To ensure data security and user authentication

1.4 Scope of the Project

The scope of the **Crypto Trading Platform using Spring Boot and React** is to develop a web-based application that allows users to securely manage and trade cryptocurrencies in a simulated environment. The system will provide a user-friendly interface where users can register, log in, and access various features related to cryptocurrency trading.

The platform will enable users to view real-time cryptocurrency market data, search for different coins, and perform buy and sell transactions. It will also include wallet management features that allow users to add balance, transfer funds between wallets, and withdraw funds to a bank account. Users will be able to monitor their investments through portfolio management tools and track all activities through a detailed transaction history.

The system will also provide strong security mechanisms such as user authentication, password recovery, and two-factor authentication to ensure safe access to the platform. Additionally, the platform will integrate external APIs to fetch real-time cryptocurrency prices and market information.

This project mainly focuses on demonstrating the implementation of a modern **full-stack web application** using technologies like **Spring Boot, React, and MySQL**, along with API integration and secure authentication. The platform will serve as an educational and demonstration tool for understanding how real-world cryptocurrency trading systems work. The scope of the project can be further extended in the future by adding advanced features such as real-time trading charts, mobile application support, **AI-based market analysis**, and blockchain wallet integration.

Out-of-Scope Features

The following functionalities are not included in the current version of the system:

- Real-time trading with actual financial transactions (live money trading)
- Integration with real banking systems for deposits and withdrawals
- Support for all global cryptocurrencies (limited coins only)
- Advanced high-frequency trading (HFT) algorithms
- Full blockchain wallet integration

CHAPTER 2: LITERATURE SURVEY / REVIEW

2.1 Overview of Existing Work

In recent years, significant research and development have been carried out in the field of cryptocurrency trading and prediction using Artificial Intelligence (AI) and Machine Learning (ML). With the increasing popularity of digital currencies such as Bitcoin and Ethereum, many researchers and developers have focused on building systems that can analyze market trends and assist in trading decisions.

Existing crypto trading platforms primarily provide features such as real-time price tracking, historical data visualization, and basic trading functionalities. These platforms rely on charts, indicators, and user interpretation for decision-making. However, they lack advanced intelligence for predicting future price movements.

Several studies have explored the use of Machine Learning algorithms such as Linear Regression, Support Vector Machines (SVM), Random Forest, and Neural Networks for cryptocurrency price prediction. Among these, Deep Learning models like Long Short-Term Memory (LSTM) have shown promising results due to their ability to handle time-series data and capture complex patterns in market trends.

Some research works also incorporate **sentiment analysis** using data from social media platforms and news articles to understand market behavior. These approaches analyze public opinion and correlate it with price fluctuations to improve prediction accuracy.

Despite these advancements, existing systems still face several limitations, such as:

- Limited prediction accuracy due to market volatility
- Lack of real-time intelligent decision support
- High computational complexity
- Difficulty in integrating multiple data sources

Therefore, there is a need for a more efficient and intelligent system that combines real-time data analysis with advanced Machine Learning techniques to provide better predictions and assist users in making informed trading decisions.

2.2 Projects, Products, or Tools Reviewed

In the field of cryptocurrency trading and analysis, several platforms and tools have been developed to assist users in managing and trading digital assets. These systems provide various features such as real-

time data visualization, trading options, and basic analytical tools. Some of the widely used platforms and tools are discussed below:

1. **CoinMarketCap:**

Provides real-time cryptocurrency prices, market capitalization, trading volume, and historical data. It is widely used for tracking market trends but does not offer trading or predictive features.

2. **Binance:**

One of the largest cryptocurrency exchanges that allows users to buy, sell, and trade digital assets. It provides advanced trading tools and charts but relies heavily on user analysis rather than AI-based predictions.

3. **Coinbase:**

A user-friendly platform designed mainly for beginners. It offers secure transactions and portfolio management but has limited advanced analytical and prediction capabilities.

4. **TradingView:**

A popular tool for technical analysis, offering advanced charts and indicators. However, it requires users to interpret data manually and does not provide automated decision-making.

5. **Kaggle:**

Provides datasets and notebooks related to cryptocurrency and machine learning. It is useful for research and model development but is not a complete trading system.

2.3 Identification of Gaps

From the study of existing cryptocurrency trading platforms and research work, several gaps and limitations have been identified. Although platforms like Binance and Coinbase provide efficient trading functionalities, they lack intelligent systems for automated decision-making and accurate predictions.

1. **Lack of AI Integration:**

Most existing platforms do not effectively utilize Artificial Intelligence and Machine Learning for price prediction.

2. **Manual Analysis Dependency:**

Users must rely on charts and indicators, which require technical knowledge and experience.

3. **Limited Prediction Accuracy:**

Available tools do not provide reliable forecasting of cryptocurrency prices like Bitcoin and Ethereum.

4. **No Automated Recommendations:**

Existing systems do not offer smart buy/sell suggestions based on data analysis.

5.Handling Large Data:

Difficulty in processing and analyzing large volumes of real-time market data efficiently.

6.User Complexity:

Platforms can be difficult for beginners due to complex interfaces and technical requirements.

- Emotional Decision-Making: Lack of automation leads to human bias and emotional trading decisions.

CHAPTER 3: SYSTEM ANALYSIS

3.1 Requirements Specification

Requirements specification defines the **functional and non-functional requirements** of the system. It helps in understanding what the system should do and how it should perform.

3.1.1 Functional Requirements

Functional requirements describe the specific operations and features that the system must perform.

1. **User Registration and Login**

Allows new users to create an account and existing users to log in securely using credentials.

2. **Real-Time Crypto Data Display**

Shows live prices and market data of cryptocurrencies like Bitcoin and Ethereum using APIs.

3. **Buy and Sell Cryptocurrency**

Enables users to perform trading operations such as buying and selling digital assets.

4. **Portfolio Management**

Helps users track their investments, total balance, and profit/loss in one place.

5. **Transaction History**

Maintains a record of all user activities like buy, sell, deposits, and withdrawals.

6. **AI/ML-Based Price Prediction**

Uses Machine Learning models to predict future price trends based on historical data.

7. **Trading Recommendations**

Provides suggestions (buy/sell/hold) based on AI analysis to support decision-making.

8. **Wallet Management**

Allows users to add balance, transfer funds, and withdraw money securely.

3.1.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance standards of the system.

1. **Performance**

The system should respond quickly and handle real-time data efficiently without delays.

2. **Security**

Ensures protection of user data using encryption, authentication, and secure APIs.

3. **Usability (User-Friendly Interface)**

The platform should be easy to use for both beginners and experienced users.

4. **Scalability**

The system should handle increasing users and large volumes of data without performance issues.

5. **Reliability**

The system should work continuously without crashes or failures.

6. **Compatibility**

The platform should work smoothly on different devices (mobile, desktop) and browsers.

3.2 Feasibility Study

A feasibility study evaluates whether the project is **practical, achievable, and beneficial** using available resources, technology, and time.

3.2.1 Technical Feasibility

Technical Feasibility

- The project uses modern technologies like **React, Node.js/Python, and Machine Learning**
- APIs provide real-time data for cryptocurrencies like Bitcoin and Ethereum
- Tools and frameworks are **easily available and widely supported**
- System can be developed using existing hardware and software

3.2.2 Economic Feasibility

- Uses **open-source technologies** (React, Python, etc.)

- No need for expensive infrastructure
- Development cost is **low and affordable**
- Maintenance cost is minimal

3.2.3 Operational Feasibility

The system is operationally feasible due to:

- System is **easy to use and user-friendly**
- Helps users in better decision-making using AI
- Reduces manual effort and improves efficiency
- Suitable for both beginners and experts

3.3 Tools and Technologies Used

The development of the **Crypto Trading Platform using Machine Learning with AI** involves a combination of modern tools and technologies for frontend, backend, database, and machine learning.

Development Environment

- **VS Code / IntelliJ IDEA:**
Code editors for development.
- **Git & GitHub:**
Version control and code management.

Frontend Technologies

- **React.js:**
Used to build a dynamic and interactive user interface.
- **Tailwind CSS / CSS:**
Used for designing responsive and attractive layouts.
- **Redux:**
Manages application state efficiently.
- **Axios:**
Handles API requests to fetch real-time data.
- **React Router DOM:**
Enables navigation between different pages.

Backend Technologies

- Node.js / Python (Flask/Django):
Used to develop server-side logic and APIs.
- REST APIs:
Enables communication between frontend and backend.

Database

MongoDB / MySQL:

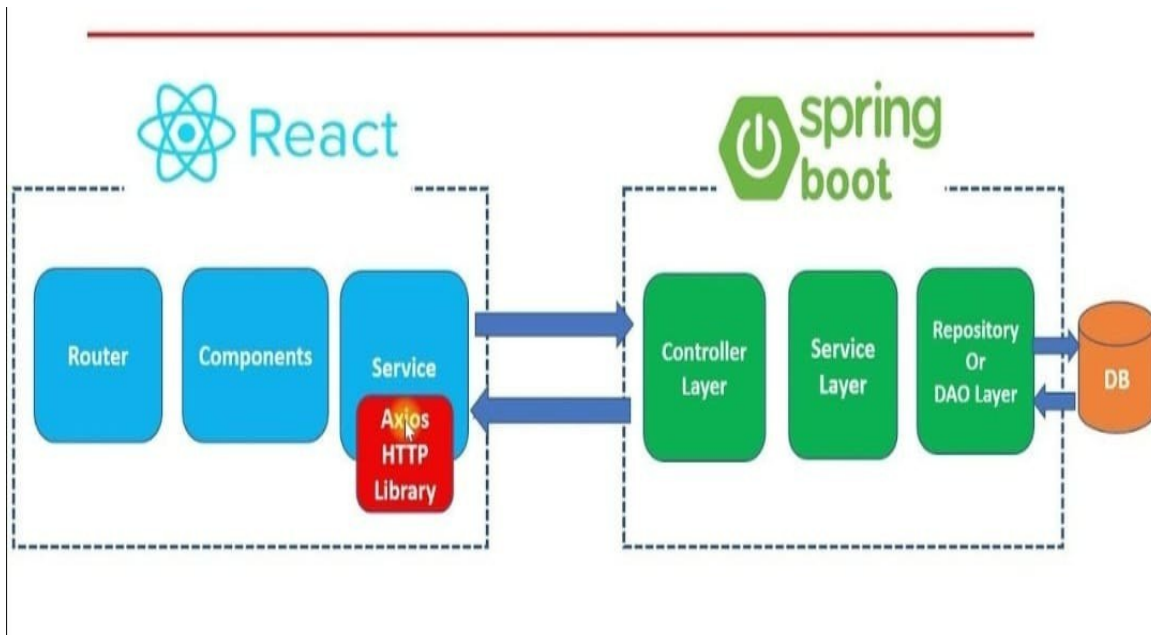
Stores user data, transaction history, and crypto information.

Additional Features

- AI Chatbot Support for answering user queries
- Real-time alerts & notifications for price changes
- Advanced charts and visualization tools
- Risk analysis and profit/loss prediction
- Two-Factor Authentication (2FA) for enhanced security
- Dark mode / customizable UI for better user experience

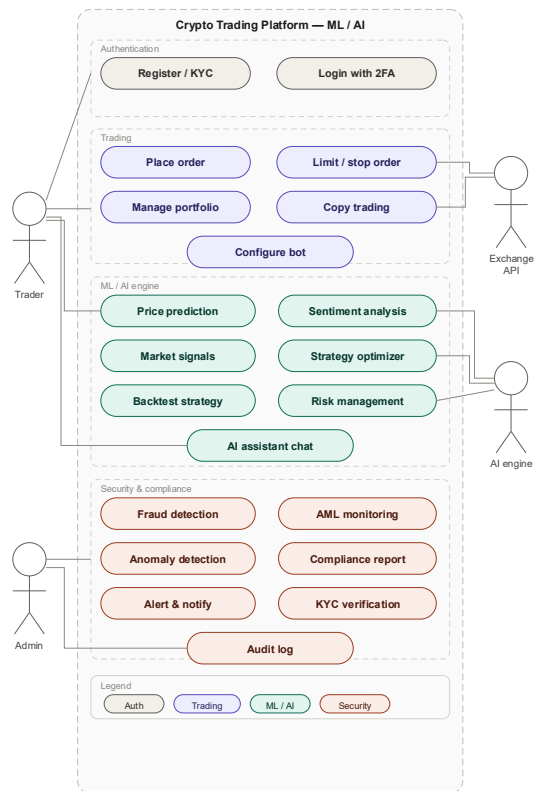
CHAPTER 4: DESIGN

4.1 System Architecture

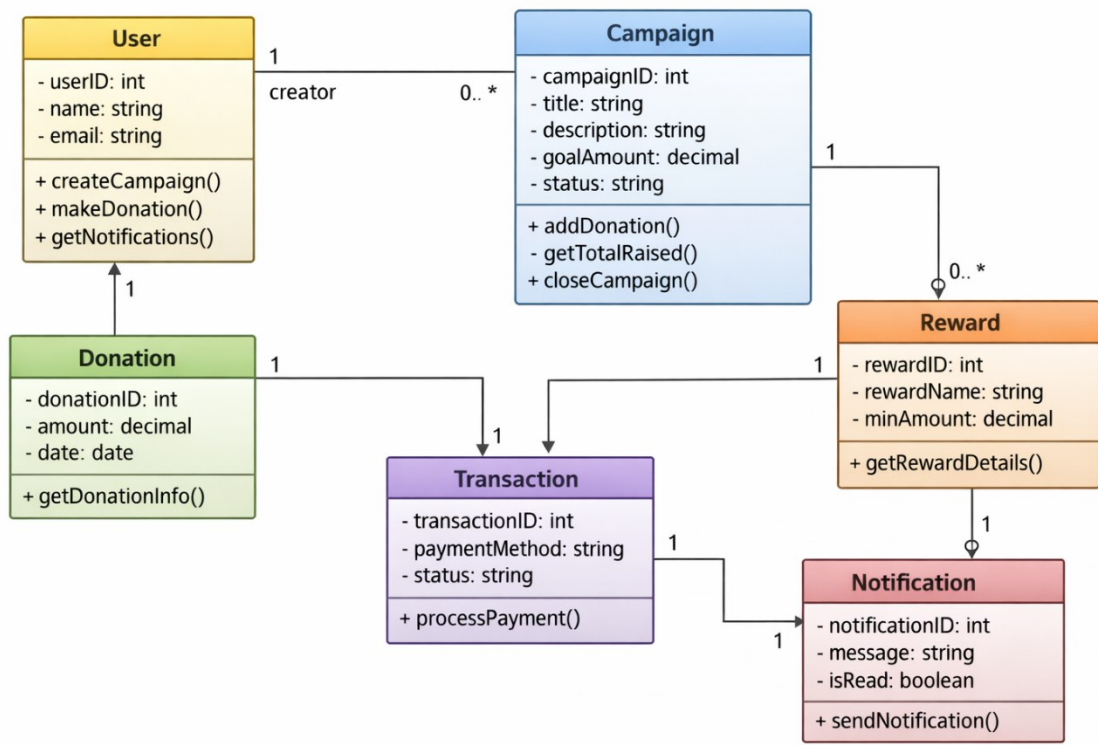


4.2 Diagrams

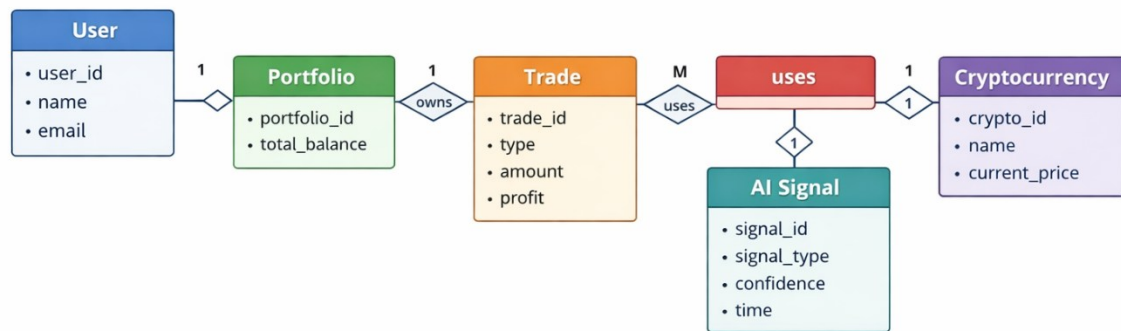
4.2.1 Use Case Diagram



4.2.2 Class Diagram

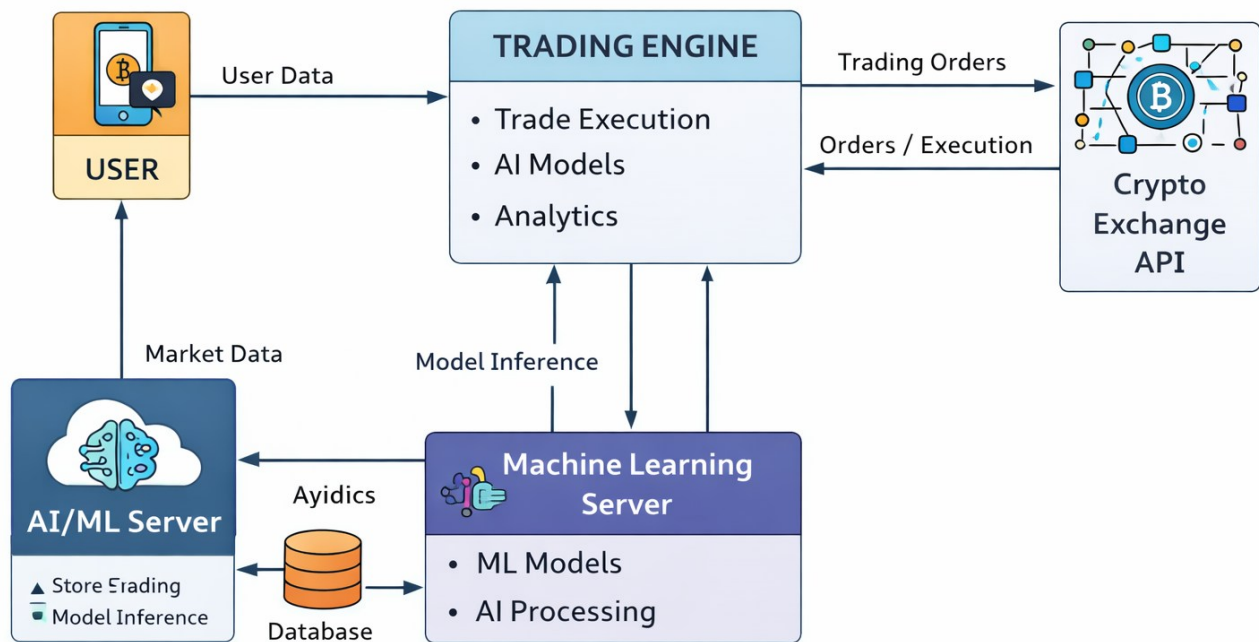


4.2.3 ER Database Diagram

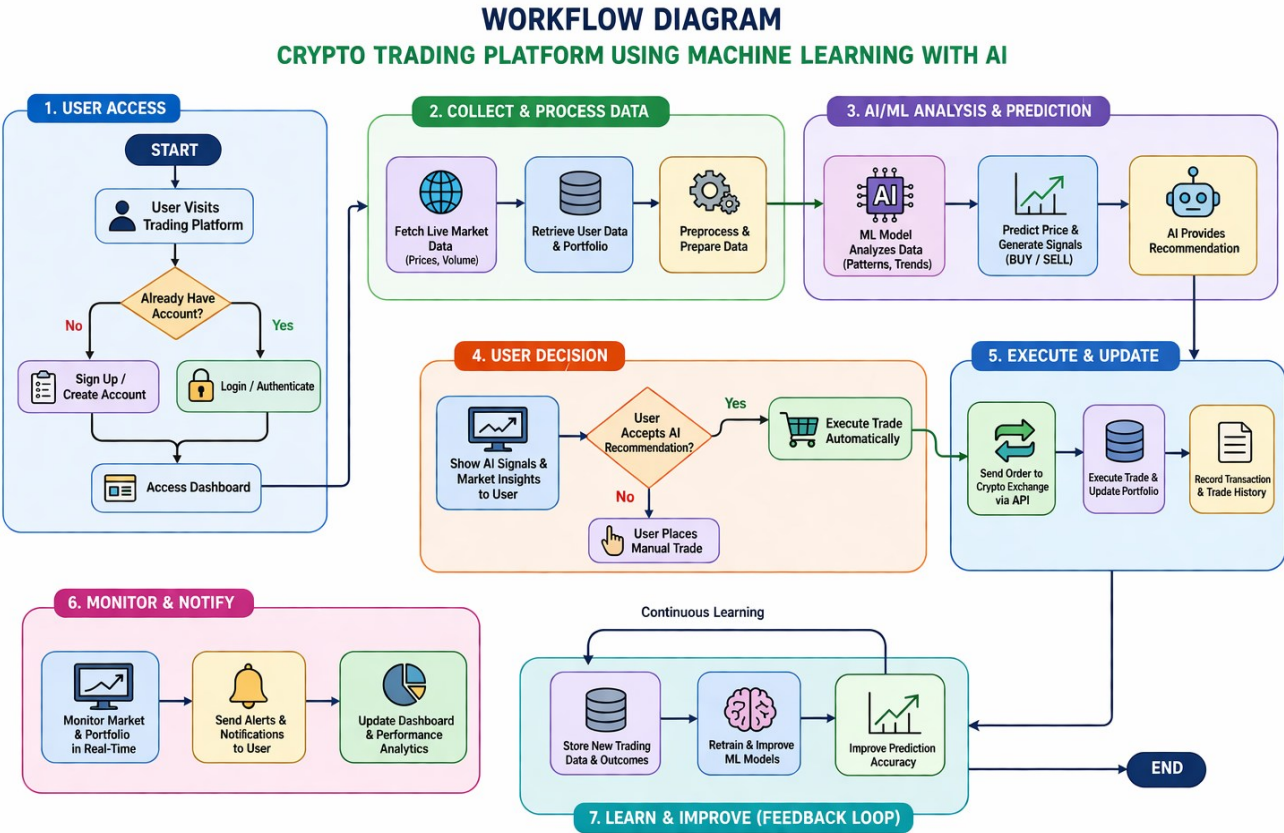


4.2.4 Block Diagram

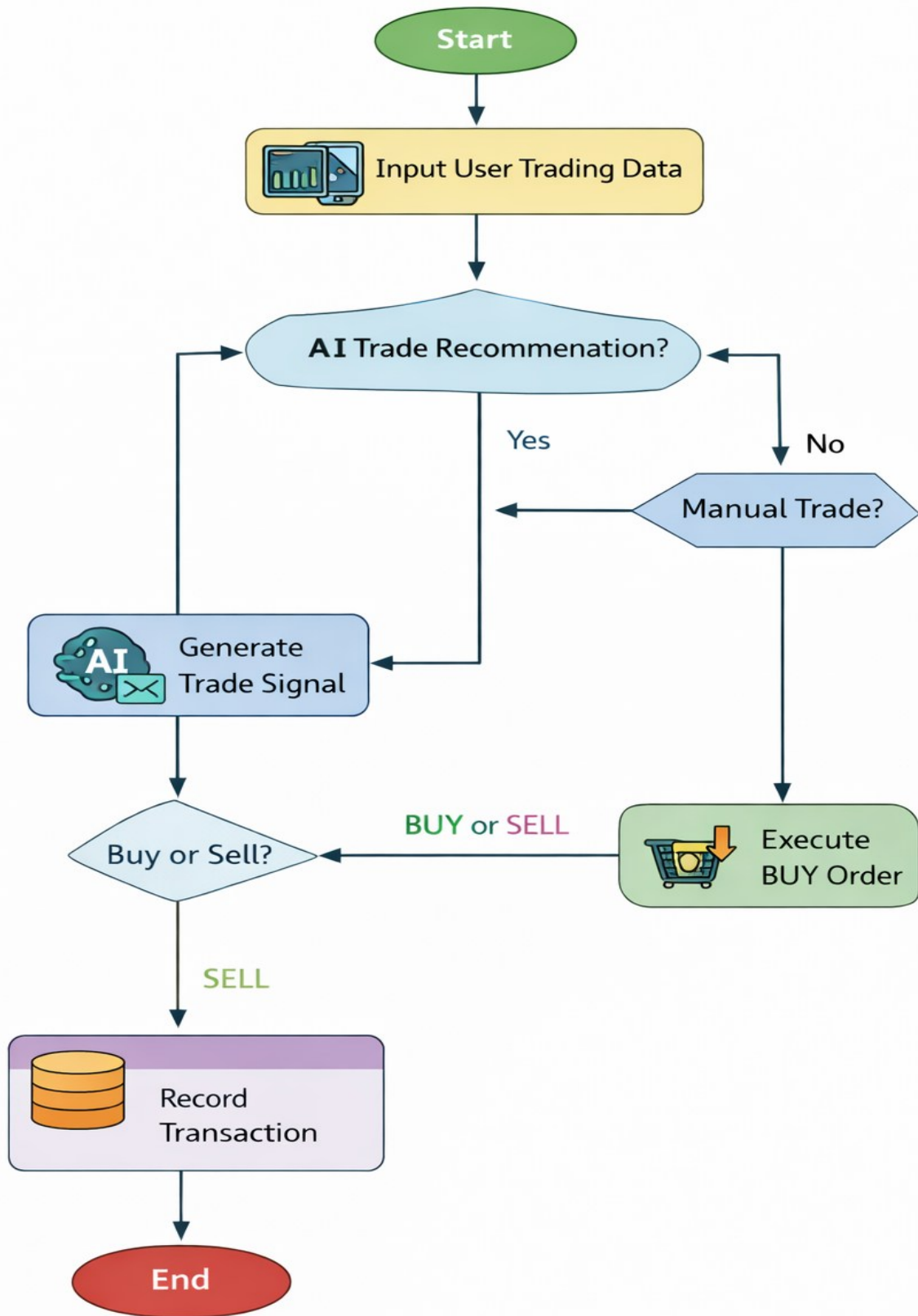
Crypto Trading Platform using Machine Learning with AI



Workflow Diagram



4.2.5 Activity Diagram



CHAPTER 5: IMPLEMENTATION

5.1 Description of Modules / Components

The **Crypto Trading Platform using Machine Learning with AI** is divided into multiple modules, each responsible for specific functionalities. These modules work together to ensure smooth operation of the system.

1. User Authentication Module

- Handles user registration and login
- Ensures secure authentication using credentials
- Manages user sessions

2. Dashboard Module

- Displays real-time cryptocurrency data
- Shows prices of coins like Bitcoin and Ethereum
- Provides charts and market trends

3. Trading Module

- Allows users to buy and sell cryptocurrencies
- Processes user transactions
- Updates user balance and portfolio

4. Portfolio Management Module

- Tracks user investments
- Displays profit/loss analysis
- Maintains wallet balance

5. Machine Learning Module

- Analyzes historical and real-time data
- Predicts future price trends
- Provides AI-based recommendations

6. API Integration Module

- Fetches real-time data from external crypto APIs
- Ensures updated market information

7. Database Module

- Stores user data, transactions, and portfolio details
- Ensures data consistency and security

8. Notification Module

- Sends alerts for price changes and trading updates
- Notifies users about important events

5.2 Frontend Development

The frontend is developed using **modern web technologies** such as **React.js, HTML, CSS, and JavaScript**. React.js is used to create reusable components and manage the dynamic behavior of the application. Tailwind CSS is used for designing a clean and responsive user interface.

Key Features:

- **User Interface (UI):**
Provides an attractive and easy-to-use interface for users.
- **Dashboard:**
Displays real-time data of cryptocurrencies like Bitcoin and Ethereum along with charts and trends.
- **Navigation System:**
Uses React Router for smooth navigation between pages such as login, dashboard, and trading.
- **API Integration:**
Axios is used to fetch real-time data from backend and external APIs.
- **Responsive Design:**
Ensures compatibility across different devices such as mobile, tablet, and desktop.
- **State Management:**
Redux is used to manage application state efficiently. The frontend ensures smooth interaction between the user and the system.

5.3 Backend Development

The backend is developed using technologies such as **Node.js / Python (Flask or Django)**. It uses **REST APIs** to communicate with the frontend and process user requests..

Core Technologies:

- **Artificial Intelligence (AI):**
Used for intelligent decision-making and automation.
- **Machine Learning (ML):**
Predicts price trends of cryptocurrencies like Bitcoin and Ethereum based on historical data.
- **React.js:**
Builds a dynamic and interactive user interface.
- **Node.js / Python:**
Handles backend logic and API development.
- **Database (MongoDB/MySQL):**
Stores user data, transactions, and portfolio details.
- **REST APIs:**
Fetch real-time cryptocurrency data and enable communication between frontend and backend.

5.4 Integration

Integration is the process of combining different modules of the system to work together as a single, unified application. In the **Crypto Trading Platform using Machine Learning with AI**, integration ensures smooth communication between the frontend, backend, database, APIs, and Machine Learning models.

The frontend, developed using React.js, interacts with the backend through REST APIs. User actions such as login, trading, and portfolio management are sent to the backend, where they are processed and stored in the database.

The backend integrates with external APIs to fetch real-time cryptocurrency data for coins like Bitcoin and Ethereum. This data is then passed to the Machine Learning module for analysis and prediction.

The Machine Learning module processes the data and returns predictions and recommendations, which are displayed on the frontend dashboard. The database stores all user-related information, including account details, transaction history, and portfolio data.

Integration Features:

- Seamless frontend and backend communication using REST APIs
- Integration with external APIs for real-time crypto data
- Machine Learning model integration for prediction and analysis
- Smooth connection between backend and database for data storage
- Real-time data synchronization across all modules
- Secure data transfer using authentication and encryption
- Efficient handling of user requests and responses

5.5 Code Snippets

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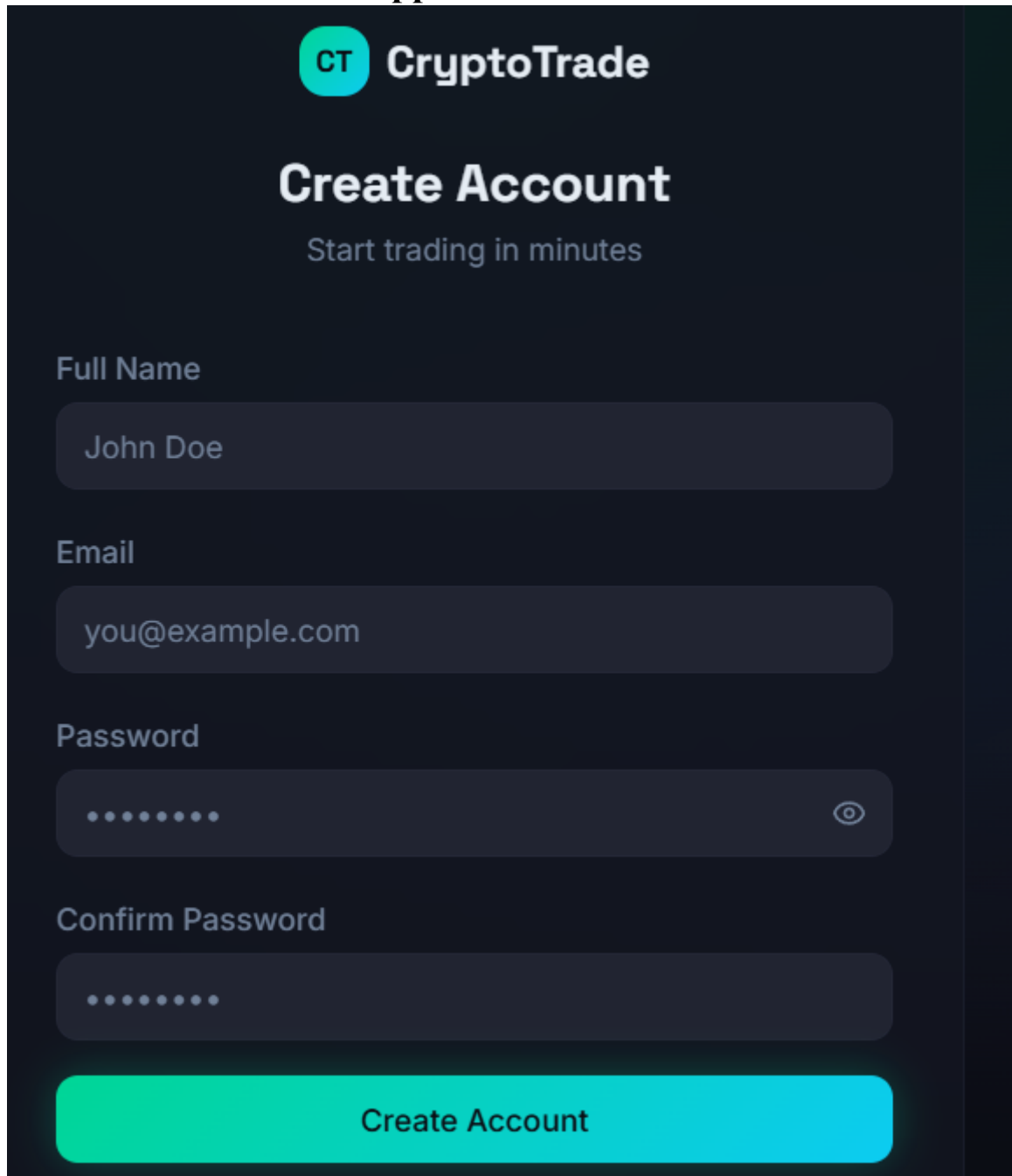
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  "integrity": "sha512-QAFe7F48e5Q3FwqTcB4BHLi3B3vPBfhH8l66sdh1k62vP8ln83vPiNCv0w8aScmGS92IwZodgOXVkV6/Rg==",
  "dev": true,
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"node_modules/@babel/runtime": {
  "version": "7.28.2",
  "resolved": "https://registry.npmjs.org/@babel/runtime/-/runtime-7.28.2.tgz",
  "integrity": "sha512-CNHjLwO08BpFwS4j6g2eTlA63TNgFvI8mFp3W12Xg1iPZDzL3JwOvW4DIHQ2JOJpL9nUdgWNL3d+ywlA==",
  "dev": true,
  "license": "MIT",
  "peer": true,
  "engines": {
    "node": ">=6.9.0"
  }
}
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    "typescript-eslint": "^8.38.0",
    "vite": "^5.4.19",
    "vitest": "^3.2.4"
  },
  "node_modules/@adobe/css-tools": {
    "version": "4.4.4",
    "resolved": "https://registry.npmjs.org/@adobe/css-tools/-/css-tools-4.4.4.tgz",
    "integrity": "sha512-Elp+iwUx5rN5+Y8xLt5/GRoG20WGoDCQ/1Fb+1LiGtvwbDavusK0jhD/eZdckHAuzcDzccnkvrEj",
    "dev": true,
    "license": "MIT"
  },
  "node_modules/@alloc/quick-lru": {
    "version": "5.2.0",
    "resolved": "https://registry.npmjs.org/@alloc/quick-lru/-/quick-lru-5.2.0.tgz",
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    "license": "MIT",
    "engines": {
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    },
    "funding": {
      "url": "https://github.com/sponsors/sindresorhus"
    }
  },
  "node_modules/@babel/code-frame": {
    "version": "7.29.0",
    "resolved": "https://registry.npmjs.org/@babel/code-frame/-/code-frame-7.29.0.tgz",
```

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"tailwind-merge": "^2.6.0",
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"vaul": "^0.9.9",
"zod": "^3.25.76"
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  "@eslint/js": "^9.32.0",
  "@playwright/test": "^1.57.0",
  "@tailwindcss/typography": "^0.5.16",
  "@testing-library/jest-dom": "^6.6.0",
  "@testing-library/react": "^16.0.0",
  "@types/node": "^22.16.5",
  "@types/react": "^18.3.23",
  "@types/react-dom": "^18.3.7",
  "@vitejs/plugin-react-swc": "^3.11.0",
  "autoprefixer": "^10.4.21",
  "eslint": "^9.32.0",
  "eslint-plugin-react-hooks": "^5.2.0",
  "eslint-plugin-react-refresh": "^0.4.20",
  "globals": "^15.15.0",
  "jsdom": "^20.0.3",
  "lovable-tagger": "^1.1.13",
  "postcss": "^8.5.6",
  "tailwindcss": "^3.4.17",
  "typescript": "^5.8.3",
  "typescript-eslint": "^8.38.0",
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"@radix-ui/react-slot": "^1.2.3",
"@radix-ui/react-switch": "^1.2.5",
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"@radix-ui/react-toast": "^1.2.14",
"@radix-ui/react-toggle": "^1.1.9",
"@radix-ui/react-toggle-group": "^1.1.10",
"@radix-ui/react-tooltip": "^1.2.7",
"@tanstack/react-query": "^5.83.0",
"class-variance-authority": "^0.7.1",
"clsx": "^2.1.1",
"cmdk": "^1.1.1",
"date-fns": "^3.6.0",
"embla-carousel-react": "^8.6.0",
"framer-motion": "^12.38.0",
"input-otp": "^1.4.2",
"lucide-react": "^0.462.0",
"next-themes": "^0.3.0",
"react": "^18.3.1",
"react-day-picker": "^8.10.1",
"react-dom": "^18.3.1",
"react-hook-form": "^7.61.1",
"react-resizable-panels": "^2.1.9",
"react-router-dom": "^6.30.1",
"recharts": "^2.15.4",
"sonner": "^1.7.4",
```

5.6 Screenshots of The Application



The screenshot displays the 'Create Account' interface for 'CryptoTrade'. The header features the 'CT' logo and the brand name. Below the title, a subtitle indicates the speed of account creation. The form consists of four input fields: 'Full Name' (containing 'John Doe'), 'Email' (containing 'you@example.com'), 'Password' (masked with dots and featuring a toggle icon), and 'Confirm Password' (also masked with dots). A prominent red 'Create Account' button is positioned at the bottom of the form.

CT CryptoTrade

Create Account

Start trading in minutes

Full Name

John Doe

Email

you@example.com

Password

..... 

Confirm Password

.....

Create Account



Welcome Back

Sign in to your account

Email

you@example.com

Password

.....



[Forgot Password?](#)

Sign In

Don't have an account? [Register](#)

Market

Crypto Market

Live prices for top cryptocurrencies

Search coins...

#	Name	Price	24h Change	Market Cap	Volume	Action
1	 Bitcoin BTC	\$67,432.50	+2.34%	\$1.32T	\$28.90B	<button>Buy</button> <button>Sell</button>
2	 Ethereum ETH	\$3,521.80	-1.12%	\$423.00B	\$15.60B	<button>Buy</button> <button>Sell</button>
3	 BNB BNB	\$594.20	+0.87%	\$91.20B	\$1.80B	<button>Buy</button> <button>Sell</button>
4	 Solana SOL	\$172.45	+5.23%	\$76.80B	\$3.20B	<button>Buy</button> <button>Sell</button>
5	 XRP XRP	\$0.6234	-0.45%	\$34.10B	\$1.40B	<button>Buy</button> <button>Sell</button>
6	 Cardano ADA	\$0.4812	+1.67%	\$17.00B	\$520.00M	<button>Buy</button> <button>Sell</button>
7	 Dogecoin DOGE	\$0.1523	-2.89%	\$21.80B	\$890.00M	<button>Buy</button> <button>Sell</button>

Wallet

Wallet

Total Balance

\$

\$128,432.50

Available

⬇

\$11,666.52

↑ Ready to trade

In Orders

⬆

\$116,765.98

Quick Actions

⬇ Add Money

⬆ Withdraw

➦ Transfer

Recent Activity

Deposit via UPI

2 days ago

+\$10,000

Withdraw to Bank

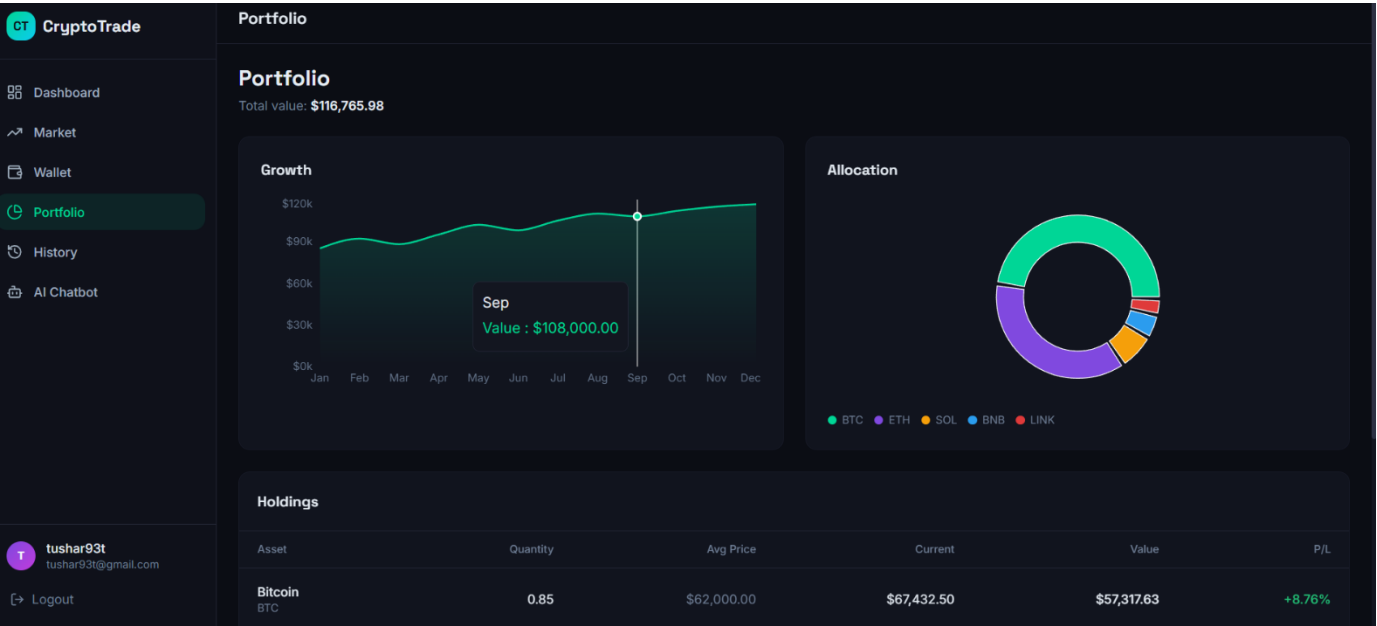
4 days ago

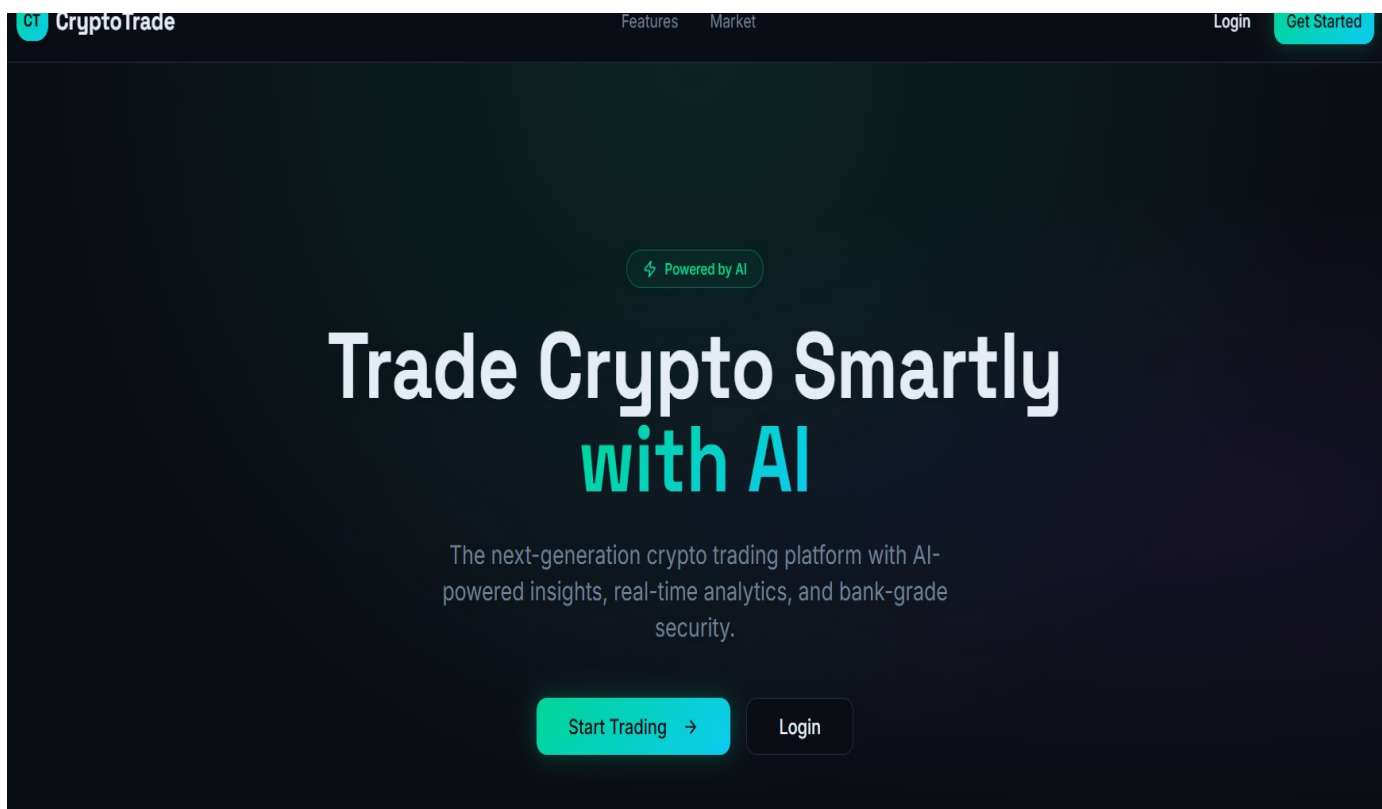
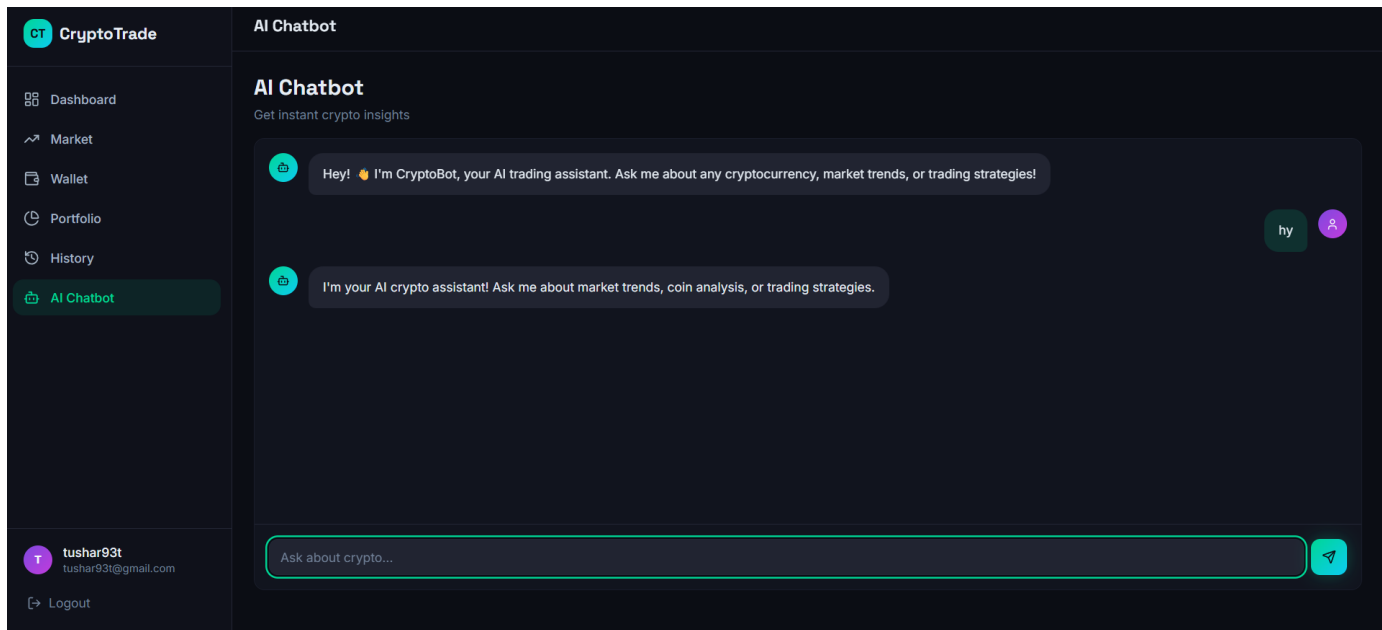
-\$2,000

Deposit via Card

1 week ago

+\$5,000





Recent Transactions			
Type	Details	Amount	Date
BUY	Bitcoin (BTC)	\$3,360.00	2026-03-28
SELL	Ethereum (ETH)	\$7,000.00	2026-03-27
DEPOSIT	—	\$10,000.00	2026-03-26
BUY	Solana (SOL)	\$2,550.00	2026-03-25
WITHDRAW	—	\$2,000.00	2026-03-24

History				
Transaction History				Deposit ▾
2 transactions				
Type	Details	Amount	Date	Status
DEPOSIT	Cash	\$10,000.00	2026-03-26	completed
DEPOSIT	Cash	\$5,000.00	2026-03-21	completed

Generated Report

CHAPTER 6: TESTING

6.1 Test Cases and Testing Methodology

Testing Approach

Testing is an important phase of the project that ensures the system works correctly, efficiently, and without errors. In the **Crypto Trading Platform using Machine Learning with AI**, different testing techniques are applied to verify each module and overall system performance.

Types of Testing Performed

- **Unit Testing:**
Each module (login, trading, ML, etc.) is tested individually.

- **Integration Testing:**
Ensures all modules work together correctly.
- **System Testing:**
Tests the complete system as a whole.
- **User Acceptance Testing (UAT):**
Verifies that the system meets user requirements. Error Handling Testing – Invalid inputs and system responses

6.1.1 Test Cases for Input Screens

Test Case ID	Input Screen	Test Scenario	Input	Expected Output	Result
TC01	Registration Form	Valid input	Correct name, email, password	Registration successful	Pass
TC02	Registration Form	Invalid email	Wrong email format	Error message displayed	Pass
TC03	Login Form	Valid login	Correct username & password	Login successful	Pass
TC04	Login Form	Invalid login	Wrong credentials	Error message shown	Pass
TC05	Trading Input	Buy crypto	Enter valid amount for Bitcoin	Transaction successful	Pass
TC06	Trading Input	Invalid amount	Negative or zero value	Error message displayed	Pass
TC07	Prediction Input	Request prediction	Select Ethereum	Shows predicted result	Pass
TC08	Empty Fields	Submit without input	Blank fields	Validation message displayed	Pass

6.2 Results of Testing

- All modules were successfully tested and performed as expected
- The system correctly handled valid and invalid inputs
- Real-time data for cryptocurrencies like Bitcoin and Ethereum was displayed accurately
- Trading operations such as buy and sell were executed without errors
- Machine Learning predictions were generated with acceptable accuracy
- Security features like authentication and validation worked properly
- The system showed stable performance under normal usage conditions.

6.3 Bug Fixes / Issues Resolved

Major Issues Identified and Fixes

Issue	Cause	Solution
Login validation error	Improper credential verification logic	Implemented secure authentication and validation
Invalid input acceptance	Missing input validation checks	Added form validation and error messages
API data delay	Slow external API response	Optimized API calls and added caching
Trading calculation error	Incorrect logic for balance updates	Fixed buy/sell transaction logic
UI display issues	Poor frontend rendering and layout bugs	Improved UI design and responsiveness
ML prediction inaccuracy	Insufficient training data and preprocessing	Enhanced model training and data cleaning
Security vulnerabilities	Weak session handling and authentication	Implemented encryption and secure sessions
Slow system performance	Unoptimized database queries and backend logic	Optimized queries and improved backend performance

CHAPTER 7: CONCLUSION & FUTURE SCOPE

7.1 Summary of the Work

The project titled “**Crypto Trading Platform using Machine Learning with AI**” was successfully designed and developed to provide an intelligent and efficient solution for cryptocurrency trading. The system integrates modern web technologies with Machine Learning algorithms to analyze market trends and assist users in making informed decisions.

The platform enables users to view real-time data of cryptocurrencies such as Bitcoin and Ethereum, perform trading operations, and manage their investment portfolios. It also incorporates AI-based prediction models that analyze historical and real-time data to forecast future price trends.

The system was developed using technologies like React for frontend, Node.js/Python for backend, and databases such as MongoDB/MySQL for data management. Proper integration of APIs ensures real-time updates, while security features protect user data and transactions.

Comprehensive testing was conducted to ensure system reliability, performance, and accuracy. All modules were verified, and identified issues were resolved successfully, resulting in a stable and efficient application.

Key Achievements

- Successfully developed an AI-based crypto trading platform
- Implemented Machine Learning models for price prediction
- Integrated real-time data APIs for cryptocurrencies like Bitcoin and Ethereum
- Enabled buy/sell trading functionality with accurate transaction handling
- Designed a user-friendly and responsive interface
- Reduced manual effort and human errors in trading decisions
- Provided smart trading recommendations using AI
- Ensured secure authentication and data protection

7.2 Limitations

Although the **Crypto Trading Platform using Machine Learning with AI** provides many advanced features, it has certain limitations:

- **Prediction Accuracy Limits:**
Machine Learning models cannot guarantee 100% accuracy due to the volatile nature of cryptocurrencies like Bitcoin and Ethereum
- **Dependence on External APIs:**
System relies on third-party APIs for real-time data, which may cause delays or downtime
- **Limited Cryptocurrency Support:**
The platform supports only a limited number of cryptocurrencies

- **No Real-Time Financial Trading (Demo-Based):**
The system may not support actual money transactions in the current version
- **Performance Constraints:**
High traffic or large datasets may affect system performance
- **Basic Security Features:**
Advanced security features like blockchain wallet integration may not be fully implemented
- **Internet Dependency:**
Requires stable internet connection for proper functioning



7.3 Future Enhancements

The **Crypto Trading Platform using Machine Learning with AI** can be further improved by adding advanced features and technologies to enhance performance, usability, and accuracy.

- Implement **advanced AI/Deep Learning models (LSTM, Neural Networks)** for better prediction accuracy
- Support a wider range of cryptocurrencies beyond Bitcoin and Ethereum
- Develop a **mobile application (Android/iOS)** for improved accessibility
- Integrate **automated trading bots** for real-time trading execution
- Add **sentiment analysis** using news and social media data
- Enhance **security features** such as blockchain wallet integration and two-factor authentication
- Improve system scalability to handle **large-scale users and data**
- Provide **multi-language support** for global users

CHAPTER 8: REFERENCES

The following books, websites, and research papers were referred to during the development of this project:

♦ Books

Tom M. Mitchell, *Machine Learning*, McGraw-Hill

Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn & TensorFlow*, O'Reilly

Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach*, Pearson

♦ Websites

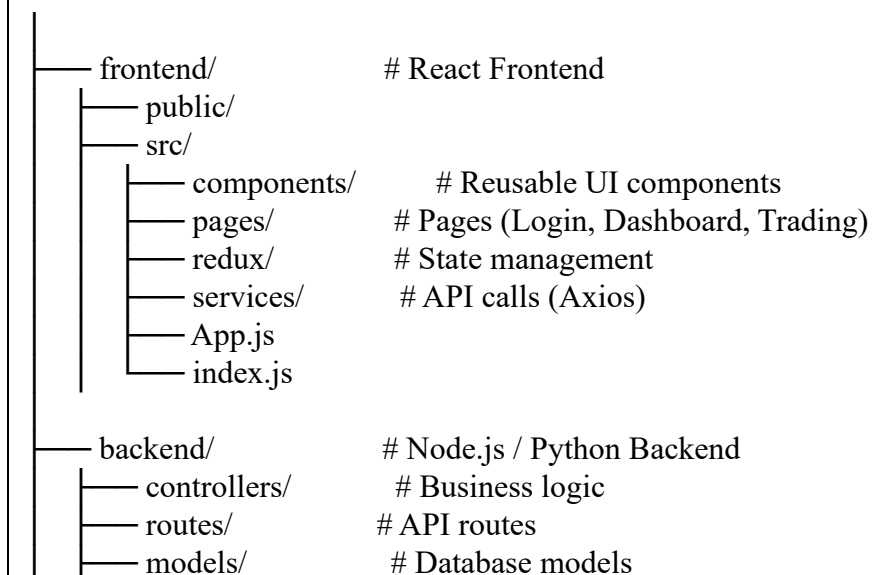
- CoinMarketCap – <https://coinmarketcap.com>
- Binance – <https://www.binance.com>

- **Coinbase** – <https://www.coinbase.com>
- **Kaggle** – <https://www.kaggle.com>
- **GitHub** – <https://github.com>

◇ **Research Papers**

- “Bitcoin Price Prediction using Machine Learning Techniques”
- “Cryptocurrency Market Analysis using Deep Learning Models”
- “AI-Based Financial Forecasting and Trading Systems”

CHAPTER 9: APPENDICES



— config/	# DB configuration
— middleware/	# Authentication & security
— server.js / app.py	
— ml-model/	# Machine Learning Module
— dataset/	# Training data
— preprocessing.py	
— model.py	
— prediction.py	
— database/	# Database scripts
— schema.sql / collections.json	
— docs/	# Documentation
— report.docx	
— diagrams/	
— .env	# Environment variables
— package.json	
— README.md	

◇ API Endpoints Table

Method	Endpoint	Description	Request Parameters	Response
POST	/api/auth/register	Register a new user	name, email, password	User created successfully
POST	/api/auth/login	User login	email, password	Authentication token

Method	Endpoint	Description	Request Parameters	Response
GET	/api/user/profile	Get user profile	token	User details
GET	/api/crypto/prices	Fetch real-time prices	coin (e.g., Bitcoin)	Current price data
GET	/api/crypto/history	Get historical data	coin, date range	Price history
POST	/api/trade/buy	Buy cryptocurrency	coin, amount	Transaction success
POST	/api/trade/sell	Sell cryptocurrency	coin, amount	Transaction success
GET	/api/portfolio	View portfolio	token	Portfolio details
GET	/api/transactions	Transaction history	token	List of transactions
POST	/api/predict	Get ML prediction	coin (e.g., Ethereum)	Predicted price

Chapter 10: Annexure -Progress Sheet

Milestone Description	Start Date	End Date	Status	Faculty Sign
Project topic selection and approval	01/02/26	07/02/26	Completed	
Requirement gathering and analysis	09/02/26	21/02/26	Completed	
System design and architecture	22/02/26	28/02/26	Completed	
Frontend development	01/3/26	06/03/26	Completed	
Backend development	06/03/26	10/03/26	Completed	

Milestone Description	Start Date	End Date	Status	Faculty Sign
API integration and database setup	11/03/26	14/03/26	Completed	
Machine Learning model implementation	15/03/26	18/03/26	Completed	
Testing (Unit & Integration)	18/03/26	25/03/26	Completed	
Documentation	25/03/26	10/04/26	Completed	

Student Signature: _____ Date: _____

Faculty Guide Signature: _____ Date: _____

HOD Signature: _____ Date: _____
